

Project Planning

Human Development Index (HDI) Prediction Using Machine Learning

Student Name: Egala Durga Rao

Department: Computer Science and Engineering (Data Science)

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1. Introduction

Project planning is an essential phase in software development that defines the activities, timeline, resources, and milestones required to successfully complete a project. Proper planning helps manage time effectively, reduce risks, and ensure that project objectives are achieved within the specified duration.

For this project, a structured plan was prepared to guide the development of the Human Development Index (HDI) Prediction System using Machine Learning and Flask.

2. Project Objectives

The major objectives of this project are:

- Develop a machine learning model for HDI prediction.
 - Analyze and preprocess the dataset.
 - Compare multiple regression algorithms.
 - Build an interactive Flask web application.
 - Provide accurate HDI predictions.
 - Create proper documentation for academic submission.
 - Upload the complete project to GitHub.
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3. Project Scope

The project focuses on predicting the Human Development Index using socio-economic indicators.

Included

- Dataset preprocessing
- Exploratory Data Analysis (EDA)
- Feature engineering
- Machine Learning model training
- Model evaluation
- Flask web application
- Responsive user interface

- [GitHub documentation](#)

Excluded

- Real-time database integration
- User authentication
- Cloud deployment
- Mobile application
- REST API development

4. Project Development Phases

The project was completed through the following phases.

Phase	Activity
Phase 1	Problem Identification
Phase 2	Requirement Analysis
Phase 3	Dataset Collection
Phase 4	Data Cleaning
Phase 5	Exploratory Data Analysis
Phase 6	Feature Selection
Phase 7	Model Training
Phase 8	Model Evaluation
Phase 9	Flask Application Development
Phase 10	Testing
Phase 11	Documentation
Phase 12	GitHub Repository Preparation

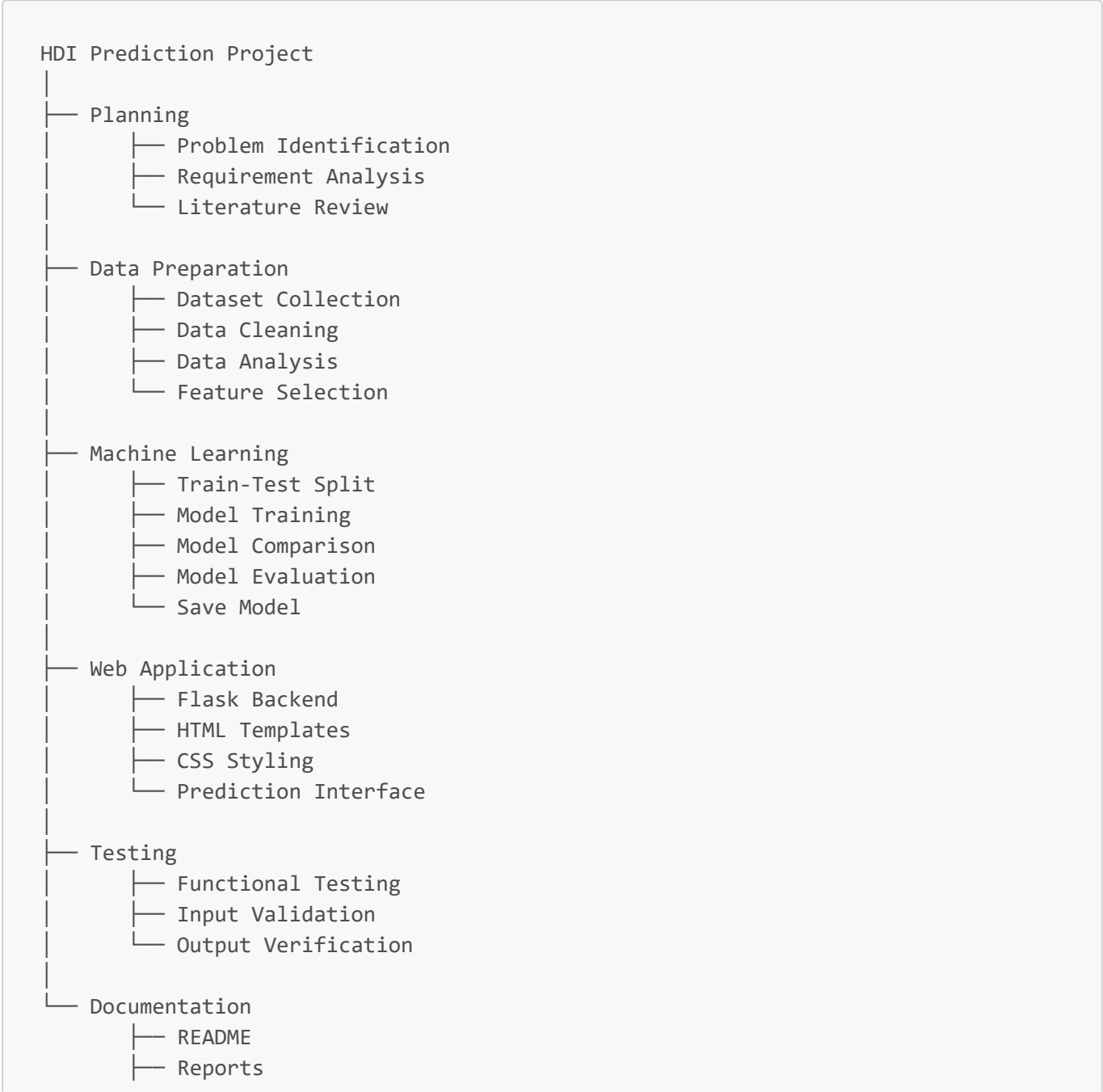
5. Project Timeline

The following timeline illustrates the planned sequence of activities.

Week	Activity
Week 1	Problem Statement and Planning
Week 2	Dataset Collection and Analysis
Week 3	Data Cleaning and Preprocessing

Week	Activity
Week 4	Exploratory Data Analysis
Week 5	Model Development
Week 6	Model Evaluation
Week 7	Flask Application Development
Week 8	UI Design and Testing
Week 9	Documentation Preparation
Week 10	GitHub Upload and Final Review

6. Work Breakdown Structure (WBS)



GitHub Repository

Final Submission

7. Resource Planning

The following resources were required.

Software Resources

- Python 3.10+
- Visual Studio Code
- Jupyter Notebook
- Flask
- Git
- GitHub

Python Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Joblib
- Flask

Hardware Resources

- Laptop/Desktop
- Minimum 8 GB RAM
- Internet Connection

8. Risk Analysis

Risk	Impact	Solution
Missing Dataset	High	Use verified public dataset
Poor Model Accuracy	High	Compare multiple algorithms
Invalid User Inputs	Medium	Apply input validation
Coding Errors	Medium	Perform testing regularly
Documentation Delay	Low	Prepare reports during development

9. Milestones

The following milestones were achieved during the project.

Milestone	Status
Project Idea Finalized	✓ Completed
Dataset Collected	✓ Completed
Data Preprocessing	✓ Completed
EDA Completed	✓ Completed
Model Training	✓ Completed
Model Evaluation	✓ Completed
Flask Application Developed	✓ Completed
User Interface Completed	✓ Completed
GitHub Repository Created	✓ Completed
Documentation Completed	✓ Completed

10. Deliverables

The final project consists of the following deliverables.

- Machine Learning Notebook
 - HDI Dataset
 - Trained Model (.pkl)
 - Flask Application
 - HTML Templates
 - CSS Stylesheet
 - README File
 - Documentation Reports
 - GitHub Repository
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11. Expected Outcomes

The project is expected to provide:

- Accurate HDI prediction.
- Practical understanding of Machine Learning.
- Experience in Flask web development.
- Knowledge of data preprocessing techniques.
- GitHub portfolio project.

- Academic project submission.
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12. Conclusion

Proper project planning played a significant role in the successful completion of the Human Development Index Prediction project. By dividing the project into well-defined phases, assigning clear objectives, and following a structured development timeline, the project was completed efficiently.

The planning process ensured that every stage—from dataset preparation to machine learning model development, Flask integration, testing, documentation, and GitHub deployment—was carried out systematically. This organized approach not only improved project quality but also enhanced learning, teamwork, and overall project management skills.